

1. MATRIX ALGEBRA

$\rho(AB) \leq \min\{\rho A, \rho B\}$; $\rho(A+B) \leq \rho A + \rho B$; $\rho(A) = \rho(A^T) = \rho(A^T A) = \rho(AA^T)$; $\mathcal{C}(A^T) = \mathcal{C}(A^T A)$; $\mathcal{N}(A) = \mathcal{N}(A^T A)$; $\mathcal{C}(A^T) = \mathcal{N}(A)^\perp$; $\text{tr}(AB) = \text{tr}(BA)$; $\text{tr} A = \sum \lambda_i$; $Ax=b$ solvable $\iff b \in \mathcal{C}(A)$; spectral $A=P\Lambda P^T$, $PP^T=P^T P=I$.
 $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ac-bd} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$; $\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1}+A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{bmatrix}$,
 $S=D-CA^{-1}B$; $(A+uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1+v^T A^{-1}u}$.

2. PROJECTIONS [24 Q1, 25 Q4]

Projection TFAE: $\mathbf{P}^2=\mathbf{P}$; $\mathcal{N}(\mathbf{P})=\mathcal{C}(I-\mathbf{P})$; $\rho(\mathbf{P})+\rho(I-\mathbf{P})=n$. **Orthogonal TFAE:** $\mathbf{P}=\mathbf{P}^T\mathbf{P}$; $\mathbf{P}^T=\mathbf{P}$ and $\mathbf{P}^2=\mathbf{P}$. Idempotent $\Rightarrow \lambda \in \{0, 1\}$

$$\text{tr}(\mathbf{P}) = \rho(\mathbf{P})$$

$$\mathcal{C}(\mathbf{P})=S, \mathcal{C}(I-\mathbf{P})=S^\perp, \mathbf{Y}^T\mathbf{P}\mathbf{Y}=\|\mathbf{P}\mathbf{Y}\|^2 \geq 0.$$

$$\mathbf{P}\mathbf{X} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{X}^T$$

$\mathbf{P}\mathbf{X}\mathbf{X}=\mathbf{X}$; $(I-\mathbf{P})\mathbf{X}=0$; $\rho(\mathbf{P}\mathbf{X})=\rho(\mathbf{X})$; invariant to $(\mathbf{X}^T\mathbf{X})^{-}$; $\mathcal{C}(\mathbf{Z}) \subseteq \mathcal{C}(\mathbf{X}) \Rightarrow \mathbf{P}\mathbf{X}\mathbf{P}\mathbf{Z}=\mathbf{P}\mathbf{Z}\mathbf{P}\mathbf{X}=\mathbf{P}\mathbf{Z}$.

$$\mathbf{X} = [\mathbf{Z} : \mathbf{W}] \Rightarrow \mathbf{P}\mathbf{X} = \mathbf{P}\mathbf{Z} + \mathbf{P}\mathbf{W}|_{\mathbf{Z}}$$

$\mathbf{P}\mathbf{W}|_{\mathbf{Z}}=M(M^T M)^{-1}M^T$, $M=(I-\mathbf{P}\mathbf{Z})\mathbf{W}$, $\rho=\rho(\mathbf{X})-\rho(\mathbf{Z})$.

[24 Q1] $\mathbf{P}, \mathbf{Q}$ symm. orth. proj.: $\mathbf{P}+\mathbf{Q}$ orth. proj. $\iff \mathbf{P}\mathbf{Q}=\mathbf{O}$; then $\rho(\mathbf{P})+\rho(\mathbf{Q})=\rho(\mathbf{P}+\mathbf{Q})$.

$\Leftarrow$: $\mathbf{P}\mathbf{Q}=0 \Rightarrow \mathbf{Q}\mathbf{P}=(\mathbf{P}\mathbf{Q})^T=0 \Rightarrow (\mathbf{P}+\mathbf{Q})^2=\mathbf{P}+\mathbf{Q}$. $\Rightarrow$: $\mathbf{P}\mathbf{Q}+\mathbf{Q}\mathbf{P}=0$; $\times \mathbf{P}$ left, $\times \mathbf{P}$ right, subtract $\Rightarrow \mathbf{P}\mathbf{Q}=\mathbf{Q}\mathbf{P} \Rightarrow 2\mathbf{P}\mathbf{Q}=0$. Rank: $\rho(\mathbf{P}+\mathbf{Q})=\text{tr}\mathbf{P}+\text{tr}\mathbf{Q}$.

[25 Q4] A, B symm., $B^2=B$, $A \succeq 0$, $I-A-B \succeq 0 \Rightarrow AB=BA=0$.

$y \in \mathcal{C}(B) \Rightarrow By=y$; $0 \leq y^T(I-A-B)y = -y^T Ay \Rightarrow y^T Ay=0$; $A=L^T L \Rightarrow \|Ly\|^2=0 \Rightarrow Ay=0$.

3. GENERALISED INVERSES [24 Q3]

$$AGA = A$$

$y \in \mathcal{C}(A) \Rightarrow A^-y$ solves $Ax=y$; all solutions $A^-y+(I-A^-A)z$. A symm.: $A^- = P\Delta P^T$, $\delta_i=1/\lambda_i$ if $\lambda_i \neq 0$, free if $\lambda_i=0$; A^T : $\delta_i=0$ (min norm); $A^T(AA^T)^-$, $(A^T A)^-A^T$ are A^- . $G=A^- \iff AG \text{ idem.}, \rho(AG)=\rho(A)$; AG projects onto $\mathcal{C}(A)$. **Invariance:** $\mathcal{R}(A) \subseteq \mathcal{R}(B)$, $\mathcal{C}(C) \subseteq \mathcal{C}(B) \Rightarrow AB^-C$ free of B^- . Reflexive $A^-AA^-=A^-$; min-norm $(A^-A)^T=A^-A$; LS $(AA^-)^T=AA^-$; A^+ : all three.

[24 Q3] $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, $A_{11} \ r \times r$, $\rho(A)=\rho(A_{11})=r \Rightarrow G = \begin{bmatrix} A_{11}^{-1} & 0 \\ 0 & 0 \end{bmatrix}$ is A^- . $AG = \begin{bmatrix} I_r & 0 \\ A_{21}A_{11}^{-1} & 0 \end{bmatrix}$, $AGA = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{21}A_{11}^{-1}A_{12} \end{bmatrix}$; $[A_{21} \ A_{22}]=M[A_{11} \ A_{12}]$, $M=A_{21}A_{11}^{-1} \Rightarrow A_{22}=A_{21}A_{11}^{-1}A_{12}$.

4. MULTIVARIATE NORMAL

$f=(2\pi)^{-n/2}|\Sigma|^{-1/2}e^{-\frac{1}{2}(y-\mu)^T\Sigma^{-1}(y-\mu)}$; MGF $e^{t^T\mu+\frac{1}{2}t^T\Sigma t}$; $\mathbf{A}\mathbf{Y} \sim N(\mathbf{A}\mu, \mathbf{A}\Sigma\mathbf{A}^T)$. $\mathbf{Y}_1|\mathbf{Y}_2 \sim N(\mu_1+\Sigma_{12}\Sigma_{22}^{-1}(\mathbf{Y}_2-\mu_2), \Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$.

$$\mathbf{A}\mathbf{Y} \perp\!\!\!\perp \mathbf{B}\mathbf{Y} \iff \mathbf{A}\Sigma\mathbf{B}^T=0; \mathbf{Y}^T\mathbf{A}\mathbf{Y} \perp\!\!\!\perp \mathbf{B}\mathbf{Y} \iff \mathbf{B}\Sigma\mathbf{A}=0$$

Moments [HW1 Q1] (A symm., $\mathbf{Y} \sim N_n(\mu, \Sigma)$)

$$\mathbb{E}(\mathbf{Y}^T\mathbf{A}\mathbf{Y}) = \text{tr}(\mathbf{A}\Sigma) + \mu^T\mathbf{A}\mu$$

$$\text{Cov}(\mathbf{Y}, \mathbf{Y}^T\mathbf{A}\mathbf{Y}) = 2\Sigma\mathbf{A}\mu$$

$$\text{Var}(\mathbf{Y}^T\mathbf{A}\mathbf{Y}) = 2\text{tr}((\mathbf{A}\Sigma)^2) + 4\mu^T\mathbf{A}\Sigma\mathbf{A}\mu$$

Distributions: χ_k^2 MGF $(1-2t)^{-k/2}$, $t < \frac{1}{2}$; $\chi_2^2 \sim \text{Exp}(\frac{1}{2})$; $\chi_k^2(\delta)$ MGF $(1-2t)^{-k/2}e^{2t\delta/(1-2t)}$, $\mathbb{E}=k+2\delta$, $\text{Var}=2k+8\delta$. $F_{a,b}=\frac{\chi_a^2/a}{\chi_b^2/b}$, $\mathbb{E}=\frac{b-2}{b-2}$; $t_\nu=\frac{N(0,1)}{\sqrt{\chi_\nu^2/\nu}}$; $t_\nu^2=F_{1,\nu}$; $\chi_a^2 \perp\!\!\!\perp \chi_b^2 \Rightarrow \frac{A}{A+B} \sim \text{Beta}(\frac{a}{2}, \frac{b}{2})$.

$$\mathbb{E}e^{t^T\mathbf{Y}^T\mathbf{A}\mathbf{Y}} = |\det(I-2tA)|^{-1/2}e^{-\frac{1}{2}t^T\mu+\frac{1}{2}t^T(I-2tA)^{-1}\mu}$$

KEY THM $\mathbf{Y} \sim N_n(\mu, I)$, A symm.:

$$\mathbf{Y}^T\mathbf{A}\mathbf{Y} \sim \chi_{\rho(A)}^2(\tfrac{1}{2}\mu^T\mathbf{A}\mu) \iff A^2 = A$$

central $\iff \mathbf{A}\mu=0$. Pf: $\prod_{j \leq r}(1-2t\lambda_j)^{-1/2}=(1-2t)^{-r/2} \forall t \iff \lambda_j=1$. $\mathbf{Y} \sim N(\mathbf{X}\beta, \sigma^2 I) \Rightarrow \text{SSE}/\sigma^2 \sim \chi_{n-\rho(\mathbf{X})}^2$, $\text{SSE} \perp\!\!\!\perp \hat{\beta}$.

FISHER-COCHRAN $\mathbf{Y} \sim N_n(\mu, I)$, A_j symm. nnd, $\mathbf{Y}^T\mathbf{Y}=\sum_j \mathbf{Y}^T A_j \mathbf{Y}$. Then $\mathbf{Y}^T A_j \mathbf{Y} \sim \chi_{\rho(A_j)}^2(\lambda_j/2)$ mutually independent

$$\iff \sum_j \rho(A_j) = n$$

and then $\lambda_j=\mu^T A_j \mu$, $\sum \lambda_j=\mu^T \mu$, $A_j^2=A_j$, $A_j A_k=0$.

5. GAUSS-MARKOV MODEL

$\mathbf{Y}=\mathbf{X}\beta+\epsilon$, $\mathbf{X} \ n \times p$, $\mathbb{E}\epsilon=0$, $\text{Var}\epsilon=\sigma^2 I$.

$$\mathbf{X}^T\mathbf{X}\beta = \mathbf{X}^T\mathbf{Y} \text{ (always solvable), } \hat{\beta} = (\mathbf{X}^T\mathbf{X})^{-}\mathbf{X}^T\mathbf{Y}$$

$\hat{\mu}=\hat{\mathbf{Y}}=\mathbf{P}\mathbf{X}\mathbf{Y}$; $e=(I-\mathbf{P}\mathbf{X})\mathbf{Y}=(I-\mathbf{P}\mathbf{X})\epsilon$; $\text{Var}(e)=\sigma^2(I-\mathbf{P}\mathbf{X})$; $\text{Var}(\hat{\mathbf{Y}})=\sigma^2\mathbf{P}\mathbf{X}$; $e \perp\!\!\!\perp \hat{\beta}$. Full rank: $\hat{\beta}=(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y} \sim N_p(\beta, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1})$; $H=\mathbf{P}\mathbf{X}$, $\sum h_{ii}=p$, $\text{Var}(e_i)=\sigma^2(1-h_{ii})$.

$$\hat{\sigma}^2 = \text{MSE} = \frac{\mathbf{Y}^T(I-\mathbf{P}\mathbf{X})\mathbf{Y}}{n-\rho(\mathbf{X})}, \quad \mathbb{E}(\text{SSE}) = \sigma^2(n-\rho(\mathbf{X}))$$

Identifiable $\iff \mathbf{X}^T\mathbf{X}$ nonsingular $\iff \rho(\mathbf{X})=p$. **ESTIMABLE** ($\exists \ell^T \mathbf{Y}$ unbiased):

$$a^T\beta \text{ est.} \iff a \in \mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T\mathbf{X}) \iff a^T z = 0 \ \forall z \in \mathcal{N}(\mathbf{X})$$

$\mathbb{E}(\ell^T \mathbf{Y})=a^T\beta \ \forall \beta \iff \ell^T \mathbf{X}=a^T \iff a=\mathbf{X}^T \ell$. Estimable $\Rightarrow a^T \hat{\beta}$ unique. **LZF** $c^T \mathbf{Y}$: $\mathbb{E}=0 \ \forall \beta \iff \mathbf{X}^T c=0$. **BLUE crit.** LUE is BLUE $\iff \text{Cov}(\ell^T \mathbf{Y}, c^T \mathbf{Y})=0$ for every LZF.

$$\mathbf{G-M:} \ a^T\beta \text{ est.} \Rightarrow a^T \hat{\beta} \text{ unique BLUE, } \text{Var} = \sigma^2 a^T (\mathbf{X}^T\mathbf{X})^{-} a$$

$a=\mathbf{X}^T\mathbf{X}u \Rightarrow a^T \hat{\beta}=u^T \mathbf{X}^T \mathbf{Y}$; $\mathbf{X}^T h=0 \Rightarrow \text{Cov}=\sigma^2 u^T \mathbf{X}^T h=0$. $\theta=\mathbf{X}^T \mu=\mathbf{X}^T \mathbf{X}\beta$ always estimable, BLUE $\mathbf{X}^T \mathbf{Y}$ [25 Q2b]. **GLS** $\text{Var}\epsilon=\Sigma \succ 0$: $\mathbf{X}^T \Sigma^{-1} \mathbf{X}\beta=\mathbf{X}^T \Sigma^{-1} \mathbf{Y}$, $\hat{\beta}^G=(\mathbf{X}^T \Sigma^{-1} \mathbf{X})^{-1} \mathbf{X}^T \Sigma^{-1} \mathbf{Y}$; $\text{Var}(a^T \hat{\beta}^G)=a^T (\mathbf{X}^T \Sigma^{-1} \mathbf{X})^{-1} a \leq \text{Var}(a^T \hat{\beta}^{LS})=a^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \Sigma \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} a$.

6. TESTING: WALD = F = LRT [25 Q1,Q3]

$\epsilon \sim N_n(0, \sigma^2 I)$, $\rho(\mathbf{X})=p$, $H_0:L^T\beta=\theta_0$, $\rho(L)=r$, $\hat{\theta}=L^T \hat{\beta}$, $V:=L^T (\mathbf{X}^T \mathbf{X})^{-1} L$, $\hat{\theta} \sim N_r(L^T \beta, \sigma^2 V)$.

$$\hat{\beta} = \hat{\beta} - (\mathbf{X}^T \mathbf{X})^{-1} L V^{-1} (\hat{\theta} - \theta_0)$$

$$\text{SSE}_{red} - \text{SSE}_{full} = (\hat{\theta} - \theta_0)^T V^{-1} (\hat{\theta} - \theta_0) = \|\mathbf{X}\hat{\beta} - \mathbf{X}\hat{\beta}\|^2$$

$$W = \frac{1}{\hat{\sigma}^2} (\hat{\theta} - \theta_0)^T V^{-1} (\hat{\theta} - \theta_0) \sim r F_{r, n-p}(\lambda/2)$$

$\lambda=\sigma^{-2}(L^T\beta-\theta_0)^T V^{-1}(L^T\beta-\theta_0)$; $\lambda=0$ under H_0 ; power $\uparrow 1$ as $\lambda \uparrow \infty$.

$$F = \frac{(\text{SSE}_{red} - \text{SSE}_{full})/(\text{df}_{red} - \text{df}_{full})}{\text{MSE}_{full}} = \frac{W}{r} \sim F_{r, n-p}$$

$\text{df}_{full}=n-p$, $\text{df}_{red}=n-p+r$; reject if $F > F_{r, n-p}^{1-\alpha}$. $r=1$: $T = \frac{\hat{\theta}-\theta_0}{\hat{\sigma}\sqrt{V}} \sim t_{n-p}$, $F=T^2$.

$$\Lambda = \left(\frac{\text{SSE}_{full}}{\text{SSE}_{red}} \right)^{n/2}, \quad -2 \log \Lambda \xrightarrow{d} \chi_r^2$$

small $\Lambda \iff \text{large } \frac{\text{SSE}_{red}-\text{SSE}_{full}}{\text{SSE}_{full}} \iff \text{large}$

F . **Sub-model** $\mathbf{X}=[\mathbf{Z}:\mathbf{W}]$, $H_0:\delta=0$: $F = \frac{\mathbf{Y}^T \mathbf{P}_{\mathbf{W}|_{\mathbf{Z}}} \mathbf{Y}/s}{\mathbf{Y}^T (I-\mathbf{P}_{\mathbf{X}}) \mathbf{Y}/(n-p)}$; $\mathbf{Y}^T \mathbf{Y}=\mathbf{Y}^T \mathbf{P}_{\mathbf{Z}} \mathbf{Y}+\mathbf{Y}^T \mathbf{P}_{\mathbf{W}|_{\mathbf{Z}}} \mathbf{Y}+\mathbf{Y}^T (I-\mathbf{P}_{\mathbf{X}}) \mathbf{Y}$, ranks $q+s+(n-p)=n$.

7. FUNDAMENTAL THEOREMS OF LS

1st $\text{SSE}=\min_{\beta} \|\mathbf{Y}-\mathbf{X}\beta\|^2=\mathbf{Y}^T(I-\mathbf{P}_{\mathbf{X}})\mathbf{Y} \sim \sigma^2 \chi_{n-\rho(\mathbf{X})}^2$.

2nd $\mathcal{C}(L) \subseteq \mathcal{C}(\mathbf{X}^T)$: $\text{SSE} \perp\!\!\!\perp (\text{SSE}_{red}-\text{SSE})$; $\text{SSE}_{red}-\text{SSE} \sim \sigma^2 \chi_m^2(\delta)$, $\delta=0 \iff L^T\beta=\theta_0$.

3rd [25 Q1] $\mathbf{X}=[\mathbf{Z}:\mathbf{W}]$, $\rho(\mathbf{X})=p$, $\rho(\mathbf{Z})=q < p$; $S_1=\min_{\beta \in \mathbb{R}^p} \|\mathbf{Y}-\mathbf{X}\beta\|^2$, $S_0=\min_{\gamma \in \mathbb{R}^q} \|\mathbf{Y}-\mathbf{Z}\gamma\|^2$; if $\mathbb{E}\mathbf{Y} \in \mathcal{C}(\mathbf{Z})$:

$$\frac{S_1}{S_0} \sim \text{Beta}\left(\frac{n-p}{2}, \frac{p-q}{2}\right)$$

$$S_1 \sim \sigma^2 \chi_{n-p}^2 \perp\!\!\!\perp (S_0-S_1) \sim \sigma^2 \chi_{p-q}^2; \frac{S_1}{S_0} = \frac{S_1}{S_1+(S_0-S_1)}.$$

8. PARTIALLING OUT [24 Q4, 25 Q3]

Frisch-Waugh $\mathbf{X}=[\mathbf{Z}:\mathbf{W}]$, $\beta=(\gamma^T, \delta^T)^T$:

$$\hat{\delta} = (\mathbf{W}^T(I-\mathbf{P}_{\mathbf{Z}})\mathbf{W})^{-1} \mathbf{W}^T(I-\mathbf{P}_{\mathbf{Z}})\mathbf{Y}$$

$$\hat{\gamma} = (\mathbf{Z}^T(I-\mathbf{P}_{\mathbf{W}})\mathbf{Z})^{-1} \mathbf{Z}^T(I-\mathbf{P}_{\mathbf{W}})\mathbf{Y}$$

$\mathbf{Z}^T \mathbf{W}=0 \Rightarrow \hat{\gamma}=(\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{Y}$ (separate fits agree).

[24 Q4] $d_j:=\mathbf{X}_{\cdot j}^T(I-\mathbf{P}_{-j})\mathbf{X}_{\cdot j}$:

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\mathbf{X}_{\cdot j}^T(I-\mathbf{P}_{-j})\mathbf{X}_{\cdot j}} = \frac{\sigma^2}{d_j}$$

$$T_j = \frac{\hat{\beta}_j}{\hat{\sigma}/\sqrt{d_j}} = \frac{1}{\hat{\sigma}/\sigma} \left(\frac{\beta_j}{\sigma} \sqrt{d_j} + W_j \right)$$

$W_j \sim N(0, 1) \perp\!\!\!\perp \hat{\sigma}^2$. Collinear $\Rightarrow (I-\mathbf{P}_{-j})\mathbf{X}_{\cdot j} \approx 0 \Rightarrow d_j \approx 0 \Rightarrow$ noncentrality $\approx 0 \Rightarrow$ no power; true 1-regressor model: $d=\|\mathbf{X}_{\cdot j}\|^2$.

$R^2 \ \mathbf{P}_1=\frac{1}{n} \mathbf{1}\mathbf{1}^T$; $\text{SST}=\mathbf{Y}^T(I-\mathbf{P}_1)\mathbf{Y}$, $\text{SSE}=\mathbf{Y}^T(I-\mathbf{P}_{\mathbf{X}})\mathbf{Y}$, $\text{SSR}=\mathbf{Y}^T(\mathbf{P}_{\mathbf{X}}-\mathbf{P}_1)\mathbf{Y}$:

$$R^2 = \frac{\mathbf{Y}^T(\mathbf{P}_{\mathbf{X}} - \mathbf{P}_1)\mathbf{Y}}{\mathbf{Y}^T(I - \mathbf{P}_1)\mathbf{Y}}, \quad R_{full}^2 - R_{sub}^2 = \frac{\mathbf{Y}^T \mathbf{P}_{\mathbf{W}|_{\mathbf{Z}}} \mathbf{Y}}{\text{SST}} \geq 0$$

Partial corr. $\rho_{YX^{(j)} \cdot \text{rest}} = \frac{(X^{(j)})^T(I-H_{(-j)})\mathbf{Y}}{\sqrt{\mathbf{Y}^T(I-H_{(-j)})\mathbf{Y}} \sqrt{(X^{(j)})^T(I-H_{(-j)})X^{(j)}}}$.

ANCOVA [25 Q3] $Y_{ij}=\mu_i+\beta_1 z_{ij}+\beta_2 w_{ij}+\epsilon_{ij}$; $S_{zz}=\sum_i \sum_j (z_{ij}-\bar{z}_i)^2$, and $S_{zw}, S_{ww}, S_{zy}, S_{wy}$ likewise; $M=\begin{bmatrix} S_{zz} & S_{zw} \\ S_{zw} & S_{ww} \end{bmatrix}$:

$$\hat{\beta} = M^{-1} \begin{bmatrix} S_{zy} \\ S_{wy} \end{bmatrix}, \quad \text{Var}(\hat{\beta}) = \sigma^2 M^{-1}, \quad \text{df} = n-r-2$$

$$H_0:\beta_2=0: U = \left(S_{ww} - \frac{S_{zw}^2}{S_{zz}} \right) \frac{\hat{\beta}_2^2}{\hat{\sigma}^2} \sim F_{1, n-r-2}$$

$$H_0:\beta_1=\beta_2=0: V = \frac{1}{\hat{\sigma}^2} \hat{\beta}^T M \hat{\beta}, \quad V/2 \sim F_{2, n-r-2}$$

9. SIMPLE LINEAR REGRESSION

$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$. **No division anywhere:**
 $S_x^2 = \sum (x_i - \bar{x})^2$, $S_Y^2 = \sum (Y_i - \bar{Y})^2$, $r_{xY} = S_{xY} / (S_x S_Y)$,
 $S_{xY} = \sum (x_i - \bar{x})(Y_i - \bar{Y}) = \sum (x_i - \bar{x})Y_i = \sum x_i Y_i - n\bar{x}\bar{Y}$.

$$\hat{\beta}_1 = \frac{S_{xY}}{S_x^2} = r_{xY} \frac{S_Y}{S_x} = \sum_i c_i Y_i, \quad c_i = \frac{x_i - \bar{x}}{S_x^2}; \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$$

$$\sum c_i = 0, \sum c_i x_i = 1, \sum c_i^2 = 1/S_x^2. \quad \mathbf{X}^T \mathbf{X} = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}, (\mathbf{X}^T \mathbf{X})^{-1} = \frac{1}{nS_x^2} \begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix},$$
$$n \sum x_i^2 - (\sum x_i)^2 = nS_x^2.$$

$$\text{Var} \hat{\beta}_1 = \frac{\sigma^2}{S_x^2}, \quad \text{Var} \hat{\beta}_0 = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_x^2} \right), \quad \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \frac{-\sigma^2 \bar{x}}{S_x^2}$$

Normal eqns $\equiv \sum e_i = 0$, $\sum e_i x_i = 0$; $\widehat{Y} = \bar{Y}$; $\text{Cov}(\bar{Y}, \hat{\beta}_1) = 0$; line passes through $(\bar{x}, \bar{Y})$; $\hat{Y}_i = \bar{Y} + \hat{\beta}_1(x_i - \bar{x})$.

Distributions ($\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$):

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_x^2}\right), \quad \hat{\beta}_0 \sim N\left(\beta_0, \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_x^2}\right)\right)$$

$$\frac{(n-2)\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-2}^2 \perp\!\!\!\perp \hat{\beta}; \quad \frac{\hat{\beta}_j - \beta_j}{s(\hat{\beta}_j)} \sim t_{n-2}$$

Any linear combination $\theta = c_0 \beta_0 + c_1 \beta_1$:

$$\text{Var}(c_0 \hat{\beta}_0 + c_1 \hat{\beta}_1) = \sigma^2 \left(\frac{c_0^2}{n} + \frac{(c_1 - c_0 \bar{x})^2}{S_x^2} \right) = \sigma^2 (c_0 \ c_1) (\mathbf{X}^T \mathbf{X})^{-1} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix}$$

$$T = \frac{c_0 \hat{\beta}_0 + c_1 \hat{\beta}_1 - v}{s(c_0 \hat{\beta}_0 + c_1 \hat{\beta}_1)} \stackrel{H_0}{\sim} t_{n-2}. \quad (c_0, c_1) = (1, x_h) \Rightarrow \text{Var}(\hat{Y}_h); \quad (0, 1) \Rightarrow \text{Var} \hat{\beta}_1. \quad \textbf{Joint}$$

$$W = \frac{1}{\hat{\sigma}^2} (\hat{\beta} - \beta)^T (\mathbf{X}^T \mathbf{X}) (\hat{\beta} - \beta) \stackrel{H_0}{\sim} 2F_{2, n-2}.$$

Hat matrix / residuals $h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_x^2}$, $\sum h_{ii} = 2$; $e = (I - H)\mathbf{Y} = (I - H)\varepsilon$,

$\text{Var}(e_i) = \sigma^2(1 - h_{ii})$, $\text{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$, $e \perp\!\!\!\perp \hat{\beta}$, $e \perp \mathbf{1}$, $e \perp x$;
 $e_i^{(-i)} = e_i / (1 - h_{ii})$.

Sums of squares

$$\text{SSE} = S_Y^2 - \hat{\beta}_1^2 S_x^2 = S_Y^2 - \frac{S_{xY}^2}{S_x^2}, \quad \text{SSR} = \hat{\beta}_1^2 S_x^2 = \frac{S_{xY}^2}{S_x^2}$$

$\text{SST} = S_Y^2 = \text{SSR} + \text{SSE}$; $\mathbb{E}(\text{SSR}) = \sigma^2 + \beta_1^2 S_x^2$; $\mathbb{E}(\text{SSE}) = (n-2)\sigma^2$; $R^2 = \text{SSR} / \text{SST} = r_{xY}^2$;
 $\hat{\sigma}^2 = \text{MSE} = \text{SSE} / (n-2)$; $\hat{\sigma}_{ML}^2 = \frac{n-2}{n} \hat{\sigma}^2$ (biased).

Source	d.f.	SS	MS
Regression	1	SSR	MSR=SSR
Error	$n-2$	SSE	MSE
Total	$n-1$	SST	

$$F = \frac{\text{MSR}}{\text{MSE}} = \frac{\text{SSR}/1}{\text{SSE}/(n-2)} \sim F_{1, n-2} = T_1^2$$

Centred form $Y_i = \beta_0^* + \beta_1(x_i - \bar{x}) + \varepsilon_i$: orthogonal design, $\hat{\beta}_0^* = \bar{Y} \perp\!\!\!\perp \hat{\beta}_1$,
 $\text{Var}(\hat{\beta}_0^*) = \sigma^2/n$, $\beta_0 = \beta_0^* - \beta_1 \bar{x}$. **No intercept** $\hat{\beta}_1 = \frac{\sum x_i Y_i}{\sum x_i^2}$, $\text{Var} = \frac{\sigma^2}{\sum x_i^2}$, only

$\sum e_i x_i = 0$, df $n-1$. **Reverse regression** $\hat{\beta}_1^{(x|Y)} = S_{xY} / S_Y^2$; $\hat{\beta}_1 \cdot \hat{\beta}_1^{(x|Y)} = r_{xY}^2$.

Prediction $\hat{Y}_h = \hat{\beta}_0 + \hat{\beta}_1 x_h$:

$$\text{Var}(\hat{Y}_h) = \sigma^2 \left(\frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_x^2} \right)$$

$$Y_{new} - \hat{Y}_h \sim N\left(0, \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_x^2} \right) \right) \perp\!\!\!\perp \text{SSE}$$

Intervals ($t^* = t_{n-2}^{1-\alpha/2}$) $\hat{\beta}_j \pm t^* s(\hat{\beta}_j)$; mean at x_h : $\hat{Y}_h \pm t^* \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_x^2}}$;

$$\text{PI: } \hat{Y}_h \pm t^* \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_x^2}}$$

Simultaneous band: replace t^* by $\sqrt{2F_{2, n-2}^{1-\alpha}}$. Ellipse:

$$(\beta - \hat{\beta})^T (\mathbf{X}^T \mathbf{X}) (\beta - \hat{\beta}) \leq 2\hat{\sigma}^2 F_{2, n-2}^{1-\alpha}; \quad \text{Bonferroni rectangle uses } t_{n-2}^{1-\alpha/4};$$

$$\mathbb{P}(A_0 \cup A_1) \leq \alpha_0 + \alpha_1.$$

Duality $C_\alpha(\mathbf{Y}) = \{\theta : \mathbf{Y} \in A_\alpha(\theta)\}$, $\mathbb{P}_\theta(\theta \in C_\alpha) \geq 1 - \alpha$.

Add/drop obs. [HW1 Q3] $\hat{\beta}_1 = \hat{\beta}_1^{(-n)} + \frac{x_n - \bar{x}}{S_x^2} e_n^{(-n)}$; $\hat{\beta}^{(-i)} = \hat{\beta} - \frac{(\mathbf{X}^T \mathbf{X})^{-1} x_i e_i}{1 - h_{ii}}$;

$$\hat{\beta}^{(n+1)} - \hat{\beta}^{(n)} = c_{n+1} (Y_{n+1} - \hat{Y}_{n+1}^{(n)}) (\mathbf{X}^T \mathbf{X})^{-1} x_{n+1}$$

$$c_{n+1} = (1 + x_{n+1}^T (\mathbf{X}^T \mathbf{X})^{-1} x_{n+1})^{-1}; \quad Y_{n+1} - \hat{Y}_{n+1}^{(n)} = (Y_{n+1} - \hat{Y}_{n+1}^{(n+1)}) / c_{n+1};$$
$$\bar{Y} = \frac{n-1}{n} \bar{Y}^{(-n)} + \frac{1}{n} Y_n, \quad \bar{x} = \frac{n-1}{n} \bar{x}^{(-n)} + \frac{1}{n} x_n.$$

10. ONE-WAY ANOVA [G — NUMERICAL]

$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij} = \mu_i + \varepsilon_{ij}$, $i \leq r$, $j \leq n_i$, $n = \sum n_i$; $\rho(\mathbf{X}) = r$ (not $r+1$); $\mathcal{N}(\mathbf{X}) = \langle (1, -1, \dots, -1)^T \rangle$. Constraint $\sum w_i \alpha_i = 0$, $\sum w_i = 1$; natural $w_i = n_i/n$:

$$\sum_i n_i \alpha_i = 0$$

Estimable: μ_i ; $\alpha_i - \alpha_{i'}$; $\sum c_i \alpha_i$ with $\sum c_i = 0$. Not: μ or α_i alone.

$$\hat{\mu}_i = \bar{Y}_{i.}, \quad \hat{\mu} = \bar{Y}.., \quad \hat{\alpha}_i = \bar{Y}_{i.} - \bar{Y}..$$

$\text{SSTO} = \sum \sum (Y_{ij} - \bar{Y}..)^2$; $\text{SSTR} = \sum_i n_i (\bar{Y}_{i.} - \bar{Y}..)^2$ (BG); $\text{SSE} = \sum \sum (Y_{ij} - \bar{Y}_{i.})^2$ (WG).

COMPUTE $T_i = \sum_j Y_{ij}$, $G = \sum_i T_i$, $\text{CF} = G^2/n$:

$$\text{SSTO} = \sum_i \sum_j Y_{ij}^2 - \text{CF}$$

$$\text{SSTR} = \sum_i T_i^2 / n_i - \text{CF}$$

$$\text{SSE} = \text{SSTO} - \text{SSTR} = \sum_i (n_i - 1) s_i^2$$

Source	d.f.	SS	MS	$\mathbb{E}(\text{MS})$
Treatments	$r-1$	SSTR	MSTR	$\sigma^2 + \frac{\sum n_i (\mu_i - \mu)^2}{r-1}$
Error	$n-r$	SSE	MSE	σ^2
Total	$n-1$	SSTO		

$$H_0: \alpha_1 = \dots = \alpha_r = 0:$$

$$F = \frac{\text{MSTR}}{\text{MSE}} = \frac{\text{SSTR}/(r-1)}{\text{SSE}/(n-r)} \sim F_{r-1, n-r}$$

reject if $F > F_{r-1, n-r}^{1-\alpha}$; under H_1 noncentral with $\lambda = \sum n_i \alpha_i^2 / \sigma^2$;

$\text{MSE} = \frac{\sum_i (n_i - 1) s_i^2}{n-r}$ (pooled). **Intervals** ($t^* = t_{n-r}^{1-\alpha/2}$, all use MSE):

$$\mu_i : \bar{Y}_{i.} \pm t^* \sqrt{\text{MSE}/n_i}$$

$$\mu_i - \mu_{i'} : \bar{Y}_{i.} - \bar{Y}_{i'}. \pm t^* \sqrt{\text{MSE} \left(\frac{1}{n_i} + \frac{1}{n_{i'}} \right)}$$

Contrast $\theta = \sum c_i \mu_i$, $\sum c_i = 0$: BLUE $\hat{\theta} = \sum c_i \bar{Y}_{i.}$, $\text{Var} = \sigma^2 \sum \frac{c_i^2}{n_i}$:

$$\sum_i c_i \bar{Y}_{i.} \pm t^* \sqrt{\text{MSE}} \sqrt{\sum_i c_i^2 / n_i}$$

$r=2$: $T_{12} = \frac{\bar{Y}_{1.} - \bar{Y}_{2.}}{\hat{\sigma} \sqrt{1/n_1 + 1/n_2}} \sim t_{n-2}$, $F = T_{12}^2$; noncentrality $\frac{\mu_1 - \mu_2}{\sigma \sqrt{1/n_1 + 1/n_2}}$.

Checks $\sum T_i = G$; $\text{SSTR} + \text{SSE} = \text{SSTO}$; $(r-1) + (n-r) = n-1$; $\sum n_i \hat{\alpha}_i = 0$; all SS ≥ 0 ; MSE to 4 d.p.

11. WORKED DESIGNS [24 Q2, 25 Q2]

[25 Q2] $Y_{ijk} = \alpha_i + \beta_j + \varepsilon_{ijk}$, $i, j \in \{1, 2\}$, $k \leq m$; rows $(1, 0, 1, 0)$, $(1, 0, 0, 1)$, $(0, 1, 1, 0)$, $(0, 1, 0, 1)$, $\rho(\mathbf{X}) = 3$:

$$\mathcal{N}(\mathbf{X}) = \langle (1, 1, -1, -1)^T \rangle \Rightarrow \text{est.} \iff a_1 + a_2 = a_3 + a_4$$

$$\hat{\mu}_{ij} = \bar{Y}_{i..} + \bar{Y}_{.j.} - \bar{Y}...; \quad \mathbf{X}^T \mathbf{Y} = (T_{1..}, T_{2..}, T_{.1.}, T_{.2.})^T.$$

- α_1 : $a^T z = 1 \neq 0 \Rightarrow$ **not estimable**.
- $\alpha_1 - \alpha_2$: BLUE $\bar{Y}_{1..} - \bar{Y}_{2..}$, $\text{Var} = \sigma^2/m$.
- $\alpha_1 + \alpha_2 + \beta_1 + \beta_2 = \mu_{11} + \mu_{22}$: BLUE $2\bar{Y}...$, $\text{Var} = \sigma^2/m$.
- $\alpha_1 - \alpha_2 + \beta_1 - \beta_2 = \mu_{11} - \mu_{22}$: BLUE $(\bar{Y}_{1..} - \bar{Y}_{2..}) + (\bar{Y}_{.1.} - \bar{Y}_{.2.})$, $\text{Var} = 2\sigma^2/m$.

[24 Q2] $Y_{1j} = \beta_1$, $Y_{2j} = \beta_1 + \beta_2$, $Y_{3j} = \beta_2 + \beta_3$, $Y_{4j} = \beta_3 + \beta_4$; rows $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$,

$\det = 1 \Rightarrow$ **full rank, all estimable**. $\hat{\beta}_1 = \bar{Y}_{1.}$, $\hat{\beta}_2 = \bar{Y}_{2.} - \bar{Y}_{1.}$,
 $\hat{\beta}_3 = \bar{Y}_{3.} - \bar{Y}_{2.} + \bar{Y}_{1.}$, $\hat{\beta}_4 = \bar{Y}_{4.} - \bar{Y}_{3.} + \bar{Y}_{2.} - \bar{Y}_{1.}$; $\text{Var} = \frac{\sigma^2}{m}, \frac{2\sigma^2}{m}, \frac{3\sigma^2}{m}, \frac{4\sigma^2}{m}$.

$$H_0: \beta_1 = \beta_2; \quad \hat{\beta}_1 - \hat{\beta}_2 = 2\bar{Y}_{1.} - \bar{Y}_{2.}, \quad \text{Var} = 5\sigma^2/m, \quad T = \frac{2\bar{Y}_{1.} - \bar{Y}_{2.}}{\hat{\sigma} \sqrt{5/m}} \sim t_{4m-4}.$$

12. EXTRAS

ANOVA completed square [G] under $\sum n_i \alpha_i = 0$:

$$G(\mu, \alpha) = \text{SSE} + n(\mu - \hat{\mu})^2 + \sum_i n_i (\alpha_i - \hat{\alpha}_i)^2$$

$\Rightarrow \hat{\mu} = \bar{Y}..$, $\hat{\alpha}_i = \bar{Y}_{i.} - \bar{Y}..$ and $\text{SSTO} = \text{SSE} + \sum n_i \hat{\alpha}_i^2$. Geometry: $(I - \mathbf{P}_1)\mathbf{Y} = (I - \mathbf{P}_\mathbf{X})\mathbf{Y} + (\mathbf{P}_\mathbf{X} - \mathbf{P}_1)\mathbf{Y}$, orthogonal.

$$\text{Var}(\hat{\alpha}_i) = \sigma^2 \left(\frac{1}{n_i} - \frac{1}{n} \right), \quad \text{Cov}(\bar{Y}_{i.}, \bar{Y}_{..}) = \frac{\sigma^2}{n}$$

Multiple regression tests $T_j = \frac{\hat{\beta}_j}{\hat{\sigma} \sqrt{[(\mathbf{X}^T \mathbf{X})^{-1}]_{jj}}} \sim t_{n-p}$; overall

$$F = \frac{\text{SSR}/(p-1)}{\text{SSE}/(n-p)} \sim F_{p-1, n-p}; \quad R_{adj}^2 = 1 - \frac{\text{SSE}/(n-p)}{\text{SST}/(n-1)}.$$

Weighted LS $\hat{\beta} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{Y}$ ($\mathbf{W} = \Sigma^{-1}$ gives GLS/BLUE).

Balanced two-way, $a \times b$, n reps $Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon$:

$\text{SSA} = bn \sum_i (\bar{Y}_{i..} - \bar{Y}...)^2$ (df $a-1$); $\text{SSB} = an \sum_j (\bar{Y}_{.j.} - \bar{Y}...)^2$ (df

$b-1$); $\text{SSAB} = n \sum_i \sum_j (\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}...)^2$ (df $(a-1)(b-1)$);

$\text{SSE} = \sum \sum \sum (Y_{ijk} - \bar{Y}_{ij.})^2$ (df $ab(n-1)$); total df $abn-1$. Additive model:

$$\hat{\mu}_{ij} = \bar{Y}_{i..} + \bar{Y}_{.j.} - \bar{Y}....$$

Misc. $\mathbb{E}(Y|X)$ minimises $\mathbb{E}[(Y - g(X))^2]$ over all g with $\mathbb{E}g^2 < \infty$. Poisson reg.: $Y|x \sim \text{Poi}(\beta_0 + \beta_1 x)$, $\mathbb{E} = \text{Var} = \beta_0 + \beta_1 x$ (heteroscedastic). Gaussian $\ell = c_n - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{X}\beta\|^2 \Rightarrow \text{MLE} = \text{LSE}$ for β ; $\hat{\sigma}_{ML}^2 = \frac{1}{n} \text{SSE}$. $Z \sim N_n(0, I)$, A symm. idem. $\Rightarrow Z^T A Z \sim \chi_{\rho(A)}^2$.